

IE 203 PS 10

Markov Chains

Q1. Random Walk

Imagine a person walking in an unusual manner. They either take a step to their left or to their right randomly with equal probability. Model this stochastic process as a Markov chain.

Q2. Gambler's Ruin

Consider two gamblers Alexei and Baboulinka playing a fair coin flipping game. Alexei has 60 kopecks and Baboulinka has 100. At each turn of the game, loser pays 5 kopecks to the winner. Game ends when one of them loses all their money. Model this stochastic process as a discrete time Markov chain.

Q3. Markov Chain Analysis

Below, a one step transition probability matrix of a discrete time Markov chain is given. Answer the questions accordingly. State indices start with 1.

$$P = \begin{pmatrix} 0 & 0.6 & 0.4 & 0 & 0 & 0 & 0 \\ 0.7 & 0 & 0.3 & 0 & 0 & 0 & 0 \\ 0.1 & 0.1 & 0 & 0.3 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 0.5 & 0.4 & 0.1 \end{pmatrix}$$

- a. Draw the state diagram.
- b. Identify the classes.
- c. Identify the types of states (recurrent, transient, absorbing).
- d. Calculate $P(X_2 = 1 \mid X_0 = 1)$.
- e. Calculate $P(X_3 = 4 \mid X_0 = 1)$.
- f. Explain the long run behavior of the Markov chain.
- g. Show that a subset of the state space is itself an ergodic and irreducible Markov chain.
- h. Calculate the steady state probabilities of the Markov chain found in part g.